PROCESS-2 corpus

400 participants

HC	200
MCI	150
Dementia	50

3 tasks, one session

- **SFT** semantic fluency (animals, 60 s)
- **PFT** phonemic fluency (letter P, 60 s)
- **CTD** picture description (Cookie Theft)

audio (16 kHz)

3 s

manual transcripts

official split

train 320 | 80 test

A The problem: the same speech is transcribed in two ways, and the way depends on diagnosis

Illustrative example (not corpus data)

What the participant says

"cat … dog, lion"

3-s silence after "cat"

Convention 1: pauses written

```
Pat: cat (3 seconds)
     dog lion
```

Convention 2: never written

```
cat dog lion
```

Sessions with convention 1

HC	52%
MCI	82%
Dem.	70%

Missing annotations are not missing pauses: HC pause as much in the audio

A text model can learn:

"seconds" → impaired

B The audit: is the gain due to the annotations or to the speech?

1 What does the model read?

← healthy impaired →

"seconds"
"second"
"buzzer"

"seconds" is the top word in 3/3 tasks and 25/25 folds

2 Remove the annotations

with / without

	all sessions	convention 1	convention 2
with	0.79	0.74	0.77
without	0.70	0.63	0.76
	−0.10	−0.11	−0.01

3 Which annotations matter?

timed pauses	−0.06
timer "buzzer"	−0.03
sighs, laughs	0.00
uncertain "(?)"	0.00

AUC lost when one category is removed: two carry it

4 Other checks

- demographics, channel **weak (AUC ≤ 0.56)**
- silence alone to audio **≈ amount of pausing**
- zero-shot LLMs **no prior use**
- cleaned transcripts **other habits remain**

C The control: a model that never reads a human annotation

Audio only	Voice activity	Speech recognition	Healthy norms	Risk score
same for everyone	pauses from audio	cat dog lion valid words / s	z by age and sex	6 visible weights

Official test set, evaluated once: AUC 0.81 [0.70, 0.91]

same under both transcription conventions; all choices made on the training set